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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Among X
Project / Solution Name	Orbyte Hub (InvoisReady) – e-Invoice & Tax Compliance Hub for Malaysian SMEs
Name (Leader)	Soh Zheng Hao
Email & Phone (Leader)	szh6875@gmail.com / +60 12-928 1791
University Affiliation(s)	Universiti Malaya × HELP University
Live Prototype URL	https://orbyte-zip.replit.app/app
Code Repository URL (public / access granted to judges)	https://github.com/szh6875-collab/SDC-Hackathon-2026-AmongX

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Orbyte Hub is a single Node.js web application. The React 19 + TypeScript front end (Vite, Tailwind CSS) runs in the browser and calls a REST API served by an Express server in the same process.

Invoices arrive as PDF, image, XML, JSON, QR code or WhatsApp-style text. The server first parses them deterministically (pdf-parse, fast-xml-parser, jsQR, Tesseract.js OCR) and, when configured, sends the text or image to Google Gemini (OpenAI or a RunPod endpoint as fallback) for structured field extraction. AI results are cross-checked against the deterministic result and routed to human approval. Invoices, company profiles, sessions and a hash-chained audit ledger are stored as JSON files on the server; ExcelJS produces .xlsm/CSV exports.

A scheduled n8n workflow calls a secured endpoint every 12 hours to fetch official LHDN e-Invoice guideline and FAQ PDFs from hasil.gov.my; extracted text, SHA-256 hashes and snapshots are saved in Supabase (PostgreSQL). The Policy Q&A retrieves saved excerpts from Supabase and asks Gemini/OpenAI for citation-checked answers.

Users may sign in with Google (Firebase Authentication) to save exports to their own Google Drive. The prototype is deployed on Replit (Replit Deployments).

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 7.0 / JavaScript on Node.js 22	Main programming languages for front end and back end	Apache-2.0 (TypeScript); MIT (Node.js)
React 19 & React DOM 19	User interface components	MIT
Vite 8 + @vitejs/plugin-react 6	Development server and production build	MIT
Tailwind CSS 4.3 (+ Autoprefixer 10)	Styling	MIT

Name & Version	Purpose in Project	License
Express 4.21 + Multer 2.5	REST API server and file uploads	MIT
@google/genai 2.4	Google Gemini SDK (invoice extraction, policy Q&A)	Apache-2.0
Firebase JS SDK 13.1 (app, auth)	Google sign-in for Google Drive export	Apache-2.0
Tesseract.js 7.0	OCR of invoice images	Apache-2.0
pdf-parse 2.4	Text extraction from invoice and guideline PDFs	Apache-2.0
fast-xml-parser 5.11	Parsing XML invoices	MIT
jsQR 1.4, qrcode 1.5, pngjs 7.0, jpeg-js 0.4	QR code decoding/generation and image decoding	Apache-2.0; MIT; MIT; BSD-3-Clause
ExcelJS 4.4, JSZip 3.10	Excel (.xlsm) / CSV / ZIP exports	MIT; MIT (dual MIT/GPL-3.0, used under MIT)
bcryptjs 3.0, dotenv 17	Password hashing; environment configuration	BSD-3-Clause; BSD-2-Clause
lucide-react 0.546, Motion 12	Icons; UI animation	ISC; MIT
tsx 4.21, esbuild 0.25	Running TypeScript server; bundling	MIT
Google Fonts: Plus Jakarta Sans, Space Mono, Young Serif	Typography	SIL Open Font License 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (Google)	Primary AI for invoice field extraction (text/image/PDF), policy summarisation and evidence-grounded Policy Q&A	Free tier (Google AI Studio key)
OpenAI API (OpenAI)	Optional fallback AI provider when Gemini quota or service fails	Paid (pay-as-you-go), optional
RunPod Serverless API – OpenAI-compatible endpoint (RunPod)	Optional alternative endpoint for invoice parsing (off by default)	Paid, optional – not used in demo
Supabase REST / RPC API (Supabase)	Saving and reading official policy snapshots, hashes and monitoring status	Free
Google Drive API v3 (Google)	Saving invoice/audit exports to the user's own Drive (drive.file scope)	Free
Firebase Authentication API (Google)	Google OAuth sign-in for Drive access	Free (Spark plan)
HASiL / LHDN and MOF official websites (public HTTPS, not an API)	Fetching official e-Invoice guidelines, FAQ PDFs and Budget speech for policy monitoring	Free (public)
Google Fonts API (Google)	Web fonts	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosting the Express + React app at https://orbyte-zip.replit.app/app	Replit plan / region

Service & Provider	Purpose in Project	Plan / Region
Supabase (managed PostgreSQL)	Policy source registry, monitored document snapshots, history and rule packages	Free plan / region
n8n	Scheduled 12-hourly policy-monitoring workflow that calls the app's secured endpoint	n8n Cloud or self-hosted / plan
Firebase Authentication (Google Cloud project)	Google sign-in for Drive integration	Spark (free)
Google Drive (user's own account)	Optional user-owned storage for exports	User's own plan
App server local disk (data/*.json)	Invoices, company profiles, sessions and hash-chained audit ledger for the demo	Same host as the app

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 3.8 Flash (gemini-3.8-flash)	Google (Gemini API)	Pre-trained (prompted with JSON schemas; primary model)	Gemini API Terms of Service
Gemini 3.1 Flash-Lite (gemini-3.1-flash-lite)	Google (Gemini API)	Pre-trained (automatic fallback on rate limit/overload)	Gemini API Terms of Service
GPT-4.1 mini (gpt-4.1-mini)	OpenAI	Pre-trained (optional cross-provider fallback)	OpenAI Terms of Use / Services Agreement
RunPod-hosted OpenAI-compatible model (set by RUNPOD_MODEL)	RunPod	Pre-trained (optional, disabled by default)	Licence of the chosen model
Tesseract LSTM English model (eng.traineddata)	Tesseract OCR project (tessdata)	Pre-trained	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
LHDN e-Invoice General & Specific Guidelines (PDF)	https://www.hasil.gov.my/e-invois/rujukan-pelaksanaan-e-invois/garis-panduan/	Public (official government)	HASiL website terms; cited with source link, not redistributed	N
LHDN e-Invoice FAQ PDFs	https://www.hasil.gov.my/e-invois/rujukan-pelaksanaan-e-invois/soalan-lazim/	Public (official government)	HASiL website terms; cited with source link	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
MOF Budget 2027 speech	https://www.mof.gov.my/portal/en/news/speech/speech-the-fifth-madani-budget-2027	Public (official government)	MOF website terms; cited with source link	N
Policy rule fixtures LHDN-v4.8 / v4.9 (JSON)	Project repository: policies/	Self-collected (team-written demo rules derived from public guidelines)	Team-owned	N
Invoice-reader benchmark (50 messages)	Project repository: benchmark/messages.json	Synthetic	Team-owned	N
Sample invoice pack (PDF/XML/JSON/QR) and demo company profiles	Project repository: sample_invoices/, data/	Synthetic	Team-owned	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Google AI Studio (Build mode, Gemini)	Scaffolding and generating application code and UI	Major
Claude Code (Anthropic, Claude Opus 5.5)	Completing this technical declaration from the project files	Minor
Other tools used, e.g. ChatGPT / Copilot	Used for	Extent

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-generated code was read and edited by team members before being merged, and checked with the repository's test scripts (tax consistency, policy Q&A retrieval, AI fallback, role/auth and compliance flows) and the 50-case reader benchmark using synthetic invoices. Checks we could not run are recorded in INTEGRATION_STATUS.md instead of being reported as passed. At runtime, AI extraction is cross-checked against deterministic parsing and needs human approval, and policy answers must cite retrieved official excerpts or the assistant declines. The team takes full responsibility for all submitted code and content.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☒ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
None	None	No official SDC sponsor-provided resource was used

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not applicable – no sponsor resource was used. The AI and cloud services in Sections 3–4 are obtained independently, and the AI layer is provider-agnostic (Gemini, OpenAI or RunPod), so any of them could be replaced.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The app processes business invoice data that users upload: supplier/buyer names, TINs, registration numbers, addresses, contact details and amounts, plus company profiles and login sessions. Uploaded files are held in server memory; extracted invoices, company data, sessions and a hash-chained audit ledger are saved as JSON files on the app server. When AI is enabled, invoice text or images are sent over HTTPS to Google Gemini (and only on failover to OpenAI or RunPod). Supabase stores only public LHDN/MOF policy text and hashes, never user data. Google Drive exports go to the user's own Drive under the limited drive.file scope. Passwords are bcrypt-hashed, sessions use random tokens that expire after 2 hours, API keys stay server-side, and data access is scoped per company. The demo uses synthetic data only; there is no automatic deletion yet, so data is kept until removed.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination / misreading: the LLM may misread amounts, TINs or dates. Mitigation: JSON-schema output, cross-checking against deterministic PDF/XML/QR/regex parsing, lower confidence on mismatches and mandatory human review. Policy Q&A: answers use only retrieved official .gov.my snapshots and approved rule packages; citations are validated and the assistant fails closed when evidence is missing or stale. Rate limits / single-provider dependency: multi-key rotation, automatic Gemini 3.8 Flash → 3.1 Flash-Lite downgrade and OpenAI/RunPod fallback; deterministic parsing still works without AI. OCR is English-only and weak on poor-quality images. SHA-256 change detection signals content changes, not legal changes, so rules are never activated automatically. Not legal or tax advice; no live MyInvois/SSM submission; demo login and local JSON storage are not production-grade.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: Shunbing

Date: 25 October 2026